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Lyapunov based PD/PID in model reference adaptive control for satellite launch vehicle systems

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ABSTRACT

This paper proposes a Lyapunov based model reference adaptive PD/PID controller for Satellite Launch Vehicle (SLV) systems. SLV is an open loop unstable plant with time varying characteristics and in general it is handled with the help of gain scheduling controllers. To improve the tracking performance, Lyapunov based adaptive PD and PID controllers are designed for model reference adaptive control structure and the performances are compared with existing gain scheduling PD/PID controller. In particular, we show that an adaptive PID controller outperforms in the presence of unmodelled dynamics and wind disturbance conditions.

1. Introduction

A Satellite Launch Vehicle (SLV) system is an aerodynamically unstable system with continuously varying parameters. Current practice is to linearize the plant along the trajectory using 'time slice' approach in each plane [1]. A classical design technique is generally used to get the suitable controllers at each of these linearized points. On the other hand, the nonlinear time varying characteristics of the plant are handled by gain scheduling and compensator switching [2]. The limitations of gain scheduling [2] are i) an open-loop adaptation scheme ii) design and validation become complex and iii) design turnaround time increases as the number and dimension of the design points increase.

The existing approaches work well for most of the SLV, however the performances are limited [1,3] due to specific assumptions. Controllers for future space transportation systems should have the capability to handle the loss of the degrees of freedom in presence of partial failure or degradation of the control effectors. Recent studies [3,4] and [5] indicate that adaptive control techniques can be used for achieving higher degrees of robustness and safety.

In reference [5], L1 adaptive output feedback architecture is proposed for a flexible crew launch vehicle to control the low frequency flexible dynamics which are closer to the rigid body dynamics. A methodology based on direct adaptive fuzzy controller is presented in [6] to control the pitch attitude dynamics of an SLV. Lyapunov principle is incorporated in the adaptation which makes sure that the system is asymptotically stable. However the selection of the semi positive definite matrix is a challenging issue.

Adaptive notch filtering technique is utilized in [7,8] and [9] to suppress the vibrations due to flexibility of a launch vehicle. In [7], it is assumed that the effect of elastic vibrations in yaw plane is negligible and the rigid body dynamics of the yaw plane is identified and used as a reference model. This model is used for the adaptation of the notch filter used for flexible mode suppression in pitch plane. This is not true for a SLV which has asymmetric pitch/yaw characteristics and flexible vibrations in yaw plane need not be negligible.

In [10], an adaptive control system based on Model Reference Adaptive Control (MRAC) is proposed which uses several basis functions to approximate the uncertainties in the system dynamics and a reference model is developed using feedback linearization. Since the vehicle dynamics and baseline controllers are time varying in nature, several linearization points and reference models are required. Choice of suitable basis functions is the biggest criticality. When higher order dynamics like flexibility and slosh are also needed to be controlled, then the number of basis functions required to represent the uncertainty increases.

Reference [11] uses an output feedback neural network adaptive element which augments an existing gain scheduled linear controller for attitude control and vibration suppression of a crew launch vehicle. In [12], a control allocation algorithm is combined

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with MRAC to guarantee the stability and tracking performance of a satellite launch vehicle system in the presence of actuator failure. An improved weighting algorithm and an anti-saturation controller were developed to compensate for the saturation error. An observer based multivariable reference adaptive control method is proposed for a wing damaged aircraft in [13].

Two different adaptive control approaches are presented for the solution of aerodynamic surface failure during formation flight in [14]. A reconfigurable control system for re-entry vehicles, based on an adaptive control strategy combined with a control allocation approach is presented in [15]. An adaptive controller based on Lyapunov stability and its application to a quadrotor UAV was presented in [16]. The adaptive controller is found to offer increased robustness to parametric uncertainties. In particular, it is found to be effective in mitigating the effects of a loss-of-thrust anomaly, which may occur due to component failure or physical damage.

The main focus of this paper is to develop a Lyapunov based adaptive PD and PID controller based on MRAC technique for a highly unstable SLV systems. The proposed study is carried out by considering only rigid body dynamics of the SLV system. The contributions of this paper are summarized as follows:

i) Lyapunov based adaptive PD controller is designed to replace the existing gain scheduled PD controller.
ii) The tracking performance is further enhanced by designing an adaptive PID controller using Lyapunov method.
iii) The results of adaptive PD/PID are compared with existing gain scheduled PD/PID controller in the presence of unmodelled dynamics and wind disturbance condition.

The modelling of SLV system is discussed in Section 2. A brief description of the gain scheduled controller, model reference adaptive controller and proposed adaptive PD and PID controller for SLV systems is presented in Section 3. In Section 4, result and discussion of different controllers are presented. Section 5 presents the concluding remarks.

2. Modelling of SLV systems

Satellite launch vehicle systems are open loop unstable systems with nonlinear time varying dynamics. Translational dynamics which are dispersions of the actual from the nominal trajectory is called as long period dynamics [1]. The system is assumed as a point mass and oscillations about the trajectory have long period compared to the oscillations about the vehicle centre of gravity. Further, long period dynamics are controlled by the launch vehicle guidance system whereas short period dynamics are being controlled by autopilot systems.

The variables appearing in Fig. 1 are defined below:

- X_I and Z_I are the axes of the inertial co-ordinate system and X_B and Z_B are the axes of the body co-ordinate system whose origin is at the centre of gravity.
- L_α is the aerodynamic load per unit angle of attack acting at the centre of pressure location. z is the normal displacement of vehicle relative to inertial frame.
- V is the forward velocity of the vehicle and $\frac{\dot{z}}{V}$ is the component of angle attack due to the lateral drift of the vehicle.
- V_W is the velocity of the wind in the lateral direction, θ is the attitude angle of the vehicle with respect to the inertial frame and δ is the angle by which the thrust vector is deflected for control purpose.
- l_c (control moment arm) is the distance from the centre of gravity to the engine gimbal point and l_α (aerodynamic moment arm) is the distance from centre of gravity of the vehicle to the centre of pressure location.

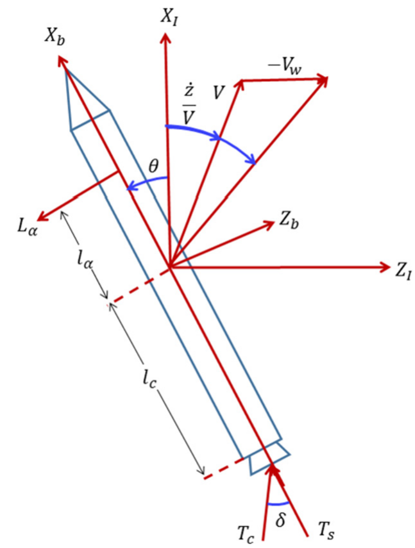

Fig. 1. Geometry of the SLV in pitch plane.

- Total thrust $T_T = T_c + T_s$, where, T_c is the control thrust and T_s is the un-deflected thrust which is not used for control.
- α is the angle of attack and α_w is the wind angle of attack. $\alpha_w = \frac{-V_w}{V}$.

For an SLV, each plane has one rotational Degree of Freedom (DoF) and two translational DoF. The translation DoF in longitudinal direction is assumed constant. i.e., the attitude control does not influence this state/DoF significantly and hence, only one lateral and rotational DoF is considered in each plane. This produces two rigid body equations – moment equation and force equation. Moment equation relates the angular acceleration ($\ddot{\theta}$) to the moments acting on the vehicle.

Force equation relates the lateral acceleration ($\ddot{z}$)/drift and the side force acting on the vehicle. Forces and torques on vehicle arise from aerodynamics (with modification from bent shape of vehicle), thrust (in the lateral direction with modification from bent shape of vehicle) and inertial forces from engine and slosh masses.

Referring to Fig. 1, equation (1) is written which consists of the force equation, moment equation and 'α' angle of attack [1].

$$\begin{aligned}
\ddot{z} &= -\frac{T_T - D}{m}\theta - \frac{L_\alpha}{m}\alpha + \frac{T_c}{m}\delta \\
\ddot{\theta} &= \mu_\alpha \alpha + \mu_c \delta \\
\alpha &= \theta + \frac{\dot{z}}{V} + \alpha_w
\end{aligned} \tag{1}$$

where,

$$\mu_\alpha = \frac{L_\alpha l_\alpha}{I}, \qquad \mu_c = \frac{T_c l_c}{I}$$

D is the drag, T is the thrust, I is the moment of inertia and m is the mass. Here thrust (T_c), control moment arm (l_c), aerodynamic force (L_α), moment arm (l_α) and inertia (I) are time varying. Hence μ_c and μ_α are time varying parameters.

Force and moment equation (1) can be written as

$$\begin{aligned}
\frac{\ddot{z}}{V} &= -\frac{(T_T - D)}{mV}\theta - \frac{L_\alpha}{mV}\alpha + \frac{T_c}{mV}\delta \\
\ddot{\theta} &= \mu_\alpha \alpha + \mu_c \delta \\
\alpha &= \theta + \frac{\dot{z}}{V} + \alpha_w
\end{aligned}$$

Neglecting the component due to wind, $\alpha = \theta + \frac{\dot{z}}{V}$.

When the aerodynamic pressure is significant, the forward velocity will be usually high. Since $\dot{z} \ll V$ and $\frac{\dot{z}}{V}$ can be neglected. Therefore $\ddot{\theta} = \mu_\alpha \theta + \mu_c \delta$.

Hence the transfer function between θ and the control input δ can be written as

$$\frac{\theta}{\delta} = \frac{\mu_c}{s^2 - \mu_\alpha} \tag{2}$$

where, μ_c, μ_α are time varying and μ_α is positive for aerodynamically unstable vehicle which gives a pole in the right half plane.

Equation (2) can be written in state space form as

$$\dot{x_p} = A_p x_p + B_p u \tag{3}$$

where,

$$A_p = \begin{bmatrix} 0 & 1 \\ \mu_\alpha & 0 \end{bmatrix}, \qquad B_p = \begin{pmatrix} 0 \\ \mu_c \end{pmatrix}$$

x_p are the states $\theta, \dot{\theta}$, u is the control input (δ).

Remark 1. The plant dynamics depend on the following time dependent parameters: Thrust of the vehicle varies with time; Mass and inertial properties (CG location and the moment of inertia) of the vehicle are continuously varying as the propellant is getting depleted; Aerodynamic force is dependent on the aerodynamic characteristics of the vehicle and the dynamic pressure which in turn depends on the velocity and altitude of the vehicle.

3. Control laws

In this paper two control laws are considered viz., gain scheduling and MRAC. These techniques are discussed subsequently.

3.1. Gain scheduling controller

In SLV, the nonlinear time varying plant is made linear time invariant using time slice approach [2]. Gain design is done at each of the linearization points and these gains are stored in the onboard computer as functions of time. In this paper, forward path gain and rate gyro path gain are scheduled with respect to time. Gain scheduling is designed using the pole placement method and closed loop poles are placed in such a way that the closed loop system performs like a second order system.

Let the control law be

$$\delta = K_A * \left(K_\theta (u_c - \theta) - K_R * \dot{\theta}\right)$$

where,

$K_A =$ forward path gain; $K_R =$ attitude rate gyro gain;

$K_\theta =$ attitude gyro gain

$\theta = x_p(1)$ and $\dot{\theta} = x_p(2)$ are the state variables

u_c is the reference signal to be tracked.

Remark 2. In order to achieve tracking of guidance commands and reduced loads due to aerodynamic forces, the control law used is of the form $\delta = K_A(-K_R\dot{\theta} + K_\theta(u_c - \theta) - K_\alpha \alpha)$. This type of control is known as load relief control. Here the tracking performance is compromised to get reduction in load and the attitude gain K_θ can be less than one. For an aerodynamically unstable vehicle, to get load relief α feedback is required. For pure attitude tracking control law α feedback is not applied and $K_\theta = 1$ gives good tracking capability as this paper considers only the tracking performance of the controller. K_A and K_R are the scheduled gains.

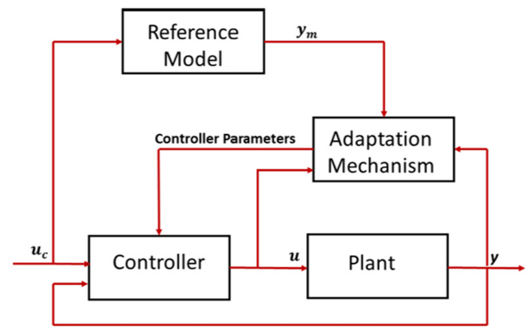

Fig. 2. Structure of MRAC.

3.2. Model reference adaptive control

In MRAC [2] the desired plant behaviour is expressed in terms of a reference model and the controller parameters are adjusted based on the feedback error between plant output and the reference model output. The mechanism for adjusting the controller parameters is obtained in two ways 1) using the gradient method (MIT rule) 2) by applying Lyapunov stability theory. In this section, a Lyapunov based MRAC as described in [17] is used to adjust the parameters of the PD and PID controller for a time varying unstable plant. The basic structure of MRAC is given in Fig. 2.

3.2.1. Adaptive PD control

A second order reference model is used and forward and feedback path gains are adaptively adjusted using the Lyapunov stability theory.

Consider the plant equation

$$\dot{x} = Ax + Bu, \quad x \in \mathcal{R}^n \tag{4}$$

where, (A, B) is controllable.

Let the reference model be given by

$$\dot{x}_m = A_m x_m + B_m u_c$$

Let the control law be

$$u = -K^* x + L^* u_c$$

The closed loop system is

$$\dot{x} = \left(A - BK^*\right)x + BL^* u_c$$

Applying the model matching condition

$$A - BK^* = A_m,$$
$$BL^* = B_m$$

The implementable control law is obtained as

$$u = -K(t)x + L(t)u_c$$

where $K(t)$, $L(t)$ are the estimates of K^*, L^*.

Tracking error $e = x - x_m$ and $\dot{e} = A_m e + B_m(-\tilde{K}x + \tilde{L}u_c)$.

Choosing the Lyapunov Function candidate

$$V(e, \tilde{K}\,\tilde{L}) = e^T Pe + tr\left(\tilde{K}^T \Gamma^{-1} \tilde{K} + \tilde{L}^T \Gamma^{-1} \tilde{L}\right)$$

where, $\tilde{K} \triangleq K - K^*$, $\tilde{L} \triangleq L - L^*$.

'P' satisfies the Lyapunov equation

$$PA_m + A_m^T P = -Q$$

$$\dot{V}(e, \tilde{K}\,\tilde{L}) = e^T \left(PA_m + A_m^T P\right)e + B_m(-\tilde{K}x + \tilde{L}u_c) + B_m(-\tilde{K}x + \tilde{L}u_c))^T Pe + 2tr\left(\tilde{K}^T \Gamma^{-1} \dot{\tilde{K}} + \tilde{L}^T \Gamma^{-1} \dot{\tilde{L}}\right)$$

$$\dot{V}(e, \tilde{K}\tilde{L}) = e^T(-Q)e + 2(-\tilde{K}x + \tilde{L}u_c))^T B_m^T Pe + 2tr\left(\tilde{K}^T \Gamma^{-1}\dot{\tilde{K}} + \tilde{L}^T \Gamma^{-1}\dot{\tilde{L}}\right)$$

To make $\dot{V} = -e^T Q e$ (negative definite)

$$\dot{\tilde{K}} = \dot{K} = \Gamma x^T B_m^T Pe,$$

$$\dot{\tilde{L}} = \dot{L} = -\Gamma u_c^T B_m^T Pe$$

Variable x_p is actually the plant state x and Γ is the adaptation gain matrix which is positive definite.

3.2.2. Adaptive PID control

Tracking performance can be improved by introducing an integral term in the controller. In the adaptive PID controller, integral gain is also adjusted using a parameter adjustment mechanism. In this formulation, the plant (4) is augmented with an integrator state and plant is the same as (1) and (2). Assuming a full state feedback control law we get

$$u = K^T x$$

Let the reference model be

$$\dot{x}_m = A_m x_m + B_m u_c$$

where, $A_m = \begin{bmatrix} 0 & 1 \\ -\omega_m^2 & -2\xi_m\omega_m \end{bmatrix}$, $B_m = \begin{bmatrix} 0 \\ \omega_m^2 \end{bmatrix}$.

Using the matching condition

$$A_p - A_m = B_p * K^T$$

To incorporate the integral control, the integral state is taken as

$$\dot{x}_I = x_p(1) - u_c$$

The plant can be augmented with the integral state as follows.

$$\begin{bmatrix} \dot{x_p} \\ \dot{x_I} \end{bmatrix} = \begin{bmatrix} A_P & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_p \\ x_I \end{bmatrix} + \begin{bmatrix} B_p \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ -1 \end{bmatrix} u_c$$

The new control law is given by

$$u = K^T x_p + K_I x_I$$

Hence the closed loop system is

$$\begin{bmatrix} \dot{x_p} \\ \dot{x_I} \end{bmatrix} = \begin{bmatrix} A_P + B_p K^T & B_p K_I \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_p \\ x_I \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} u_c$$

Applying additional matching condition we get

$$\begin{bmatrix} A_P + B_p K^T & B_p K_I \\ 1 & 0 \end{bmatrix} = \overline{A_m}$$

$$\text{where } \overline{A_m} = \begin{bmatrix} 0 & 1 & 0 \\ a1 & a2 & a3 \\ 1 & 0 & 0 \end{bmatrix}$$

The parameter update law can be obtained using the Lyapunov function based approach and is given by

$$\begin{bmatrix} \dot{K} \\ \dot{K_I} \end{bmatrix} = -\Gamma \begin{bmatrix} x_p \\ x_I \end{bmatrix} e^T P B_p$$

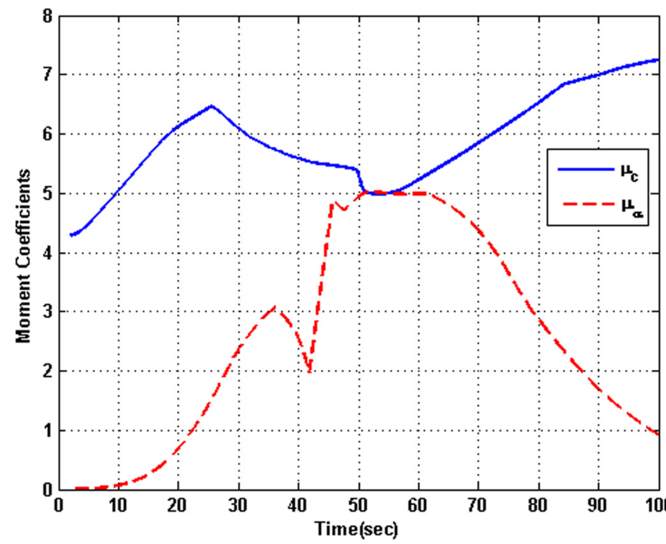

Fig. 3. The plant characteristic.

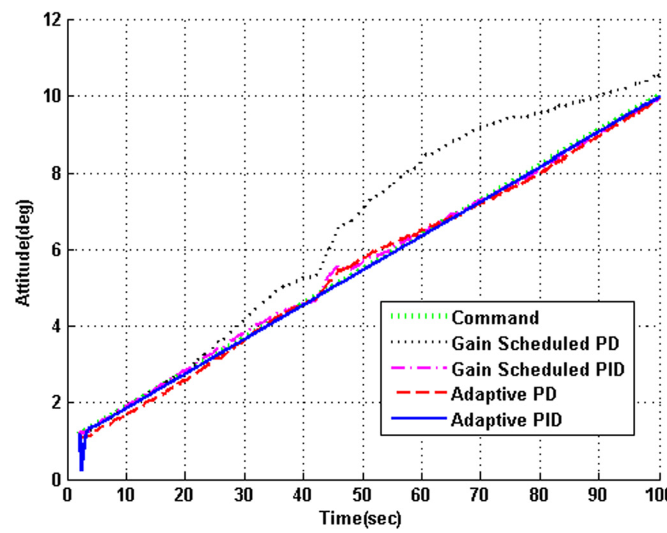

Fig. 4. Tracking Performance of the controllers for a ramp command.

4. Results and discussions

The SLV system considered is time varying and the control moment coefficient shown in the numerator of (2) which is dependent on the thrust and inertial properties of the vehicle. The two poles of the plant $\pm\sqrt{\mu_\alpha}$ depend on the aerodynamic characteristics and the inertial properties. μ_α is an indication of the aerodynamic instability of the plant. Under closed loop control, the plant is to follow a desired attitude command during the atmospheric phase given by the guidance commands. The variation of the plant parameters (μ_c and μ_α) for a typical launch vehicle during the atmospheric phase is given in Fig. 3. The plant is simulated for duration of 100 s with various controllers in the loop and the performances are compared. The maximum limit for control command is 8 deg. The parameters of the reference model is chosen as $\omega_m = 3.5$ rad/s and $\xi_m = 0.7$.

4.1. Performance Analysis with different commands

Two different command signals are used to compare the performance of the gain scheduled controllers and the proposed adaptive controllers. First a ramp command is set and then studies are re-

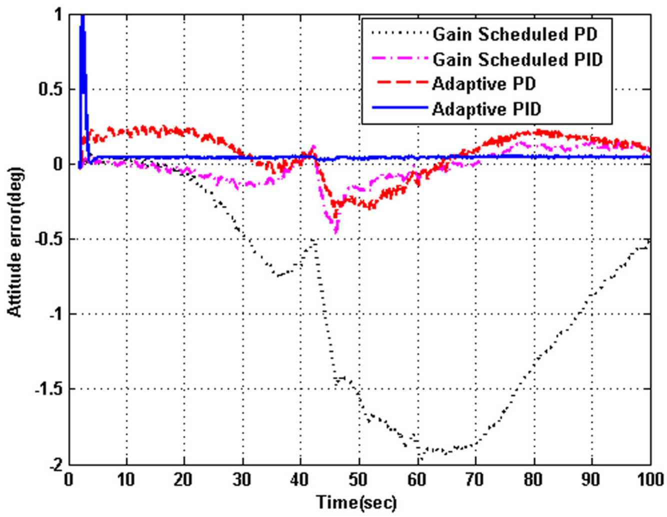

Fig. 5. Tracking error of the controllers for a ramp command.

peated with a typical open loop guidance command. The following assumptions are applied in the simulation studies:

1. Gaussian white noise is introduced in the sensor output for full scale.
2. Actuator dynamics was not considered in the design phase. However, a second order actuator model with acceleration, slew rate and position limits is used in simulations.

The tracking performance of the controllers and corresponding tracking error for a ramp command is shown in Fig. 4 and Fig. 5. From Fig. 5 it is evident that the proposed adaptive PID controller produces a better tracking performance (less error) compared to adaptive PD and gain scheduled PD/PID controllers. However, it is noted that the time taken to capture the initial conditions is slightly more in case of adaptive PID controller. This introduces more control commands during initial capture for adaptive PID controller and produces least tracking error when time progresses. During transonic regime (from 35 s to 45 s) where the aerodynamic parameters are changing drastically (refer Fig. 3), adaptive PID gives less error compared to other controllers.

The output response of the controllers is shown in Fig. 6 and it is seen that the control command saturates at 8 degrees for the adaptive PID controller and the gain scheduled PD controller for about 0.5 s and 2.5 s respectively. It is also noted that control signals for adaptive PD/PID controller are more noisy compared to gain scheduled controllers.

Launch vehicles generally follow a predefined ground computed attitude steering during atmospheric flight and subsequently use closed loop guidance algorithm for on-board steering computation till mission completes. Hence the simulations are repeated with a typical open loop guidance command where aggressive commands will come to reduce the angle of attack build up due to wind.

Tracking performance of different controllers for a typical guidance command is shown in Fig. 7. The capture performance of different controllers during the initial manoeuvre (zoomed view of Fig. 7) is shown in Fig. 8. During the initial manoeuvres where step like commands are present, both adaptive PID and gain scheduled PID show slightly sluggish response and introduces small error build up as shown in Fig. 9. However adaptive PID improves the performance in increasing timescale and provides a near zero tracking error during the steady phase.

From Fig. 10, it is seen that the adaptive PID control requires maximum control demand than other controllers during initial capture. The control response of the adaptive PD/PID controllers

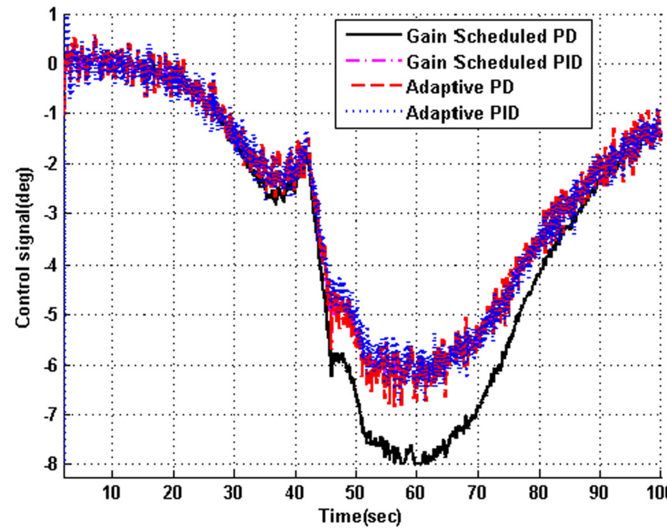

Fig. 6. Controller's response for a ramp command.

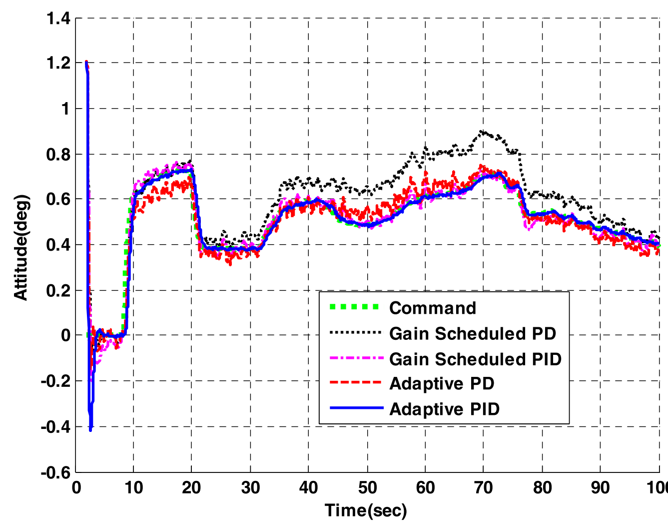

Fig. 7. Tracking Performance of the controllers for typical guidance command.

are more noisy compared to that of the gain scheduled controllers which is similar to ramp command.

Table 1 shows the tracking performance comparison of different controllers for a typical guidance commands for the SLV systems. It is evident that adaptive PID controller provides better tracking ability and almost zero integral absolute square error compared with adaptive PD and gains scheduled PD/PID controllers however it requires more control demand during initial capture. It is also observed that the tracking 'errors' introduced by the gain scheduled PID controller and adaptive PD controller are the same, however adaptive PD controller produces less integral absolute square error.

4.2. Robustness verification

4.2.1. Perturbed plant condition

The control design is performed based on the nominal plant parameters which are obtained from wind tunnel tests/Computational Fluid Dynamics simulations. During the transonic regime (40–70 s) maximum deviation is obtained from the nominal data. Hence robustness verification is performed in simulation by perturbing the plant parameters in the particular region which is shown in Fig. 11. The tracking performance of the adaptive PID is best (almost zero error) compared to other controllers shown in Fig. 12.

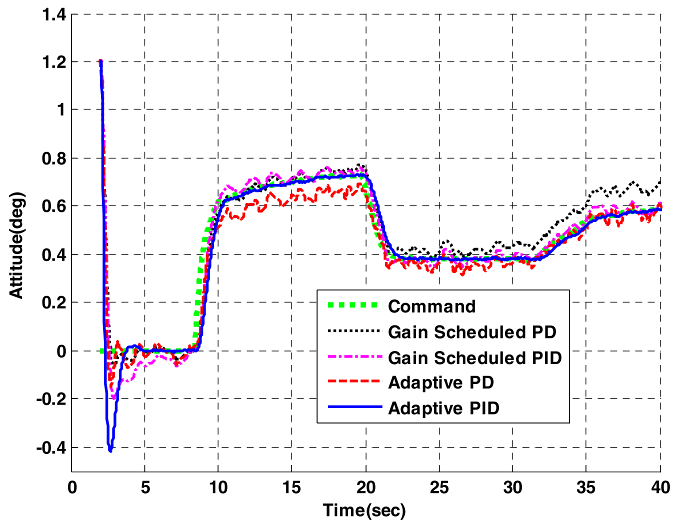

Fig. 8. Zoomed view of Fig. 7 (0–40 s).

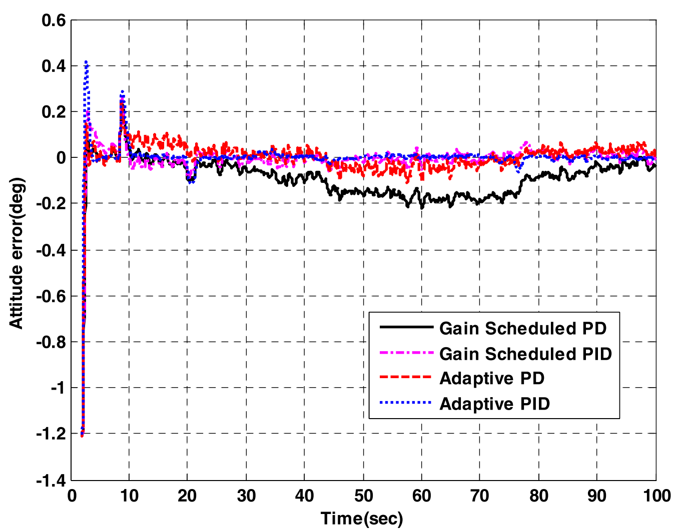

Fig. 9. Tracking Error of different Controllers for typical guidance command.

Table 2 shows the comparison of performance in terms of tracking error and control demand for adaptive PD/PID and gain scheduled PD/PID with typical guidance command system under parameter perturbations. It is observed that the adaptive PID control works magnificently with near zero tracking error and requires slightly increased control demand for the perturbed system compared to the nominal system discussed in section 4.1.

4.2.2. Wind disturbance condition

During the atmospheric flight, a SLV experiences severe disturbances like wind gusts, shear etc. This causes the angle of attack build up which in turn causes increase in vehicle loads (i.e. product of dynamic pressure and angle of attack). In general, load relief control laws are implemented during the high dynamic pressure regime to reduce the angle of attack build up due to wind disturbance. To assess the wind disturbance rejection capabilities of the different controllers, a synthetic wind profile is injected to the plant keeping the command at zero. The angle of attack build up due to wind disturbance is almost same in all controllers as seen in Fig. 13. The attitude error build up due to wind gust is more for gain scheduled PD and adaptive PID shows the least error build up as shown in Fig. 14.

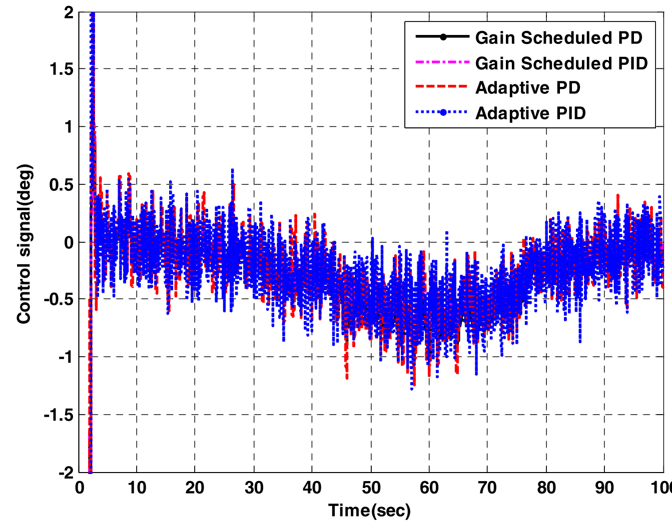

Fig. 10. Controller response for typical guidance command.

The wind disturbance rejection capability of different controllers is shown in Table 3. It is evident that adaptive PID controller provides least angle of attack build up and almost zero integral absolute square error compared with gain scheduled controllers and adaptive PD controller. However it requires more control demand during initial capture.

5. Conclusions

This paper considered an adaptive PD/PID controller to stabilize an open loop unstable SLV system during the atmospheric phase of the flight. The Lyapunov based adaptive PD/PID controller for MRAC structure was designed and simulated with ramp, typical guidance commands and results were compared with conventional PD/PID gain scheduled controller. It was shown that both adaptive PD/PID controllers were giving better tracking performance in terms of least tracking error compared to gain scheduled PD/PID controller. Interestingly it was found that an adaptive PID controller provides more robust to parametric perturbations and wind disturbances as compared to adaptive PD controller and existing gain scheduling PD/PID controllers.

Table 1
Comparison of the performance for different schemes with typical guidance commands.

Scheme	Tracking Error (deg)	Maximum Control Demand (deg)	Integral Absolute Square Error (deg)
Gain Scheduled PD Controller	0.2206	4.097 deg for initial capture 1.0 deg during high disturbance	1.5965
Gain Scheduled PID controller	0.07	4.168 deg for initial capture 0.8733 deg during high disturbance	0.5305
Adaptive PD Controller	0.07	6 deg for initial capture 1.245 deg during high disturbance	0.1748
Adaptive PID Controller	0.0015	8 deg for initial capture 1.279 deg during high disturbance	0.0287

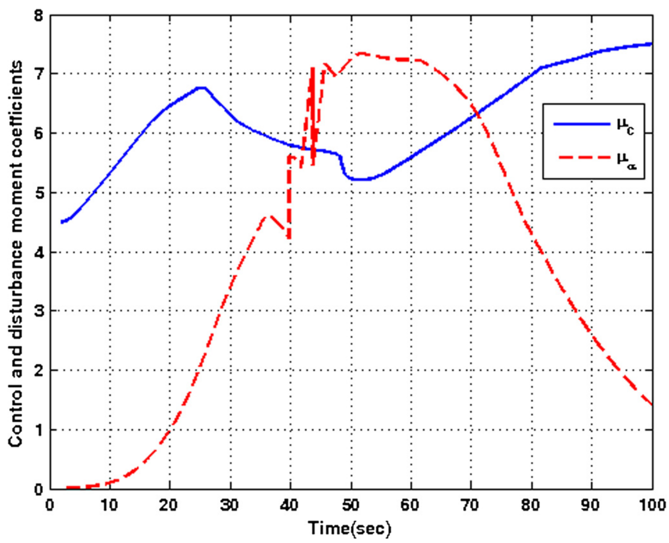

Fig. 11. Perturbed Plant.

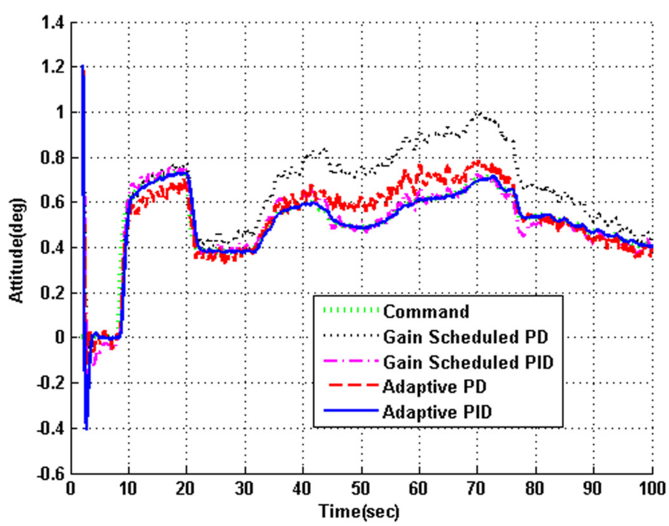

Fig. 12. Tracking Performance of the controllers under parameter perturbations.

The present study considered only the rigid body dynamics of the SLV system. The development of new control schemes to reduce the effect of high frequency dynamics like slosh and flexibility along with position and rate limited actuator on rigid body is an interesting future direction.

Conflict of interest statement

No conflict of interest.

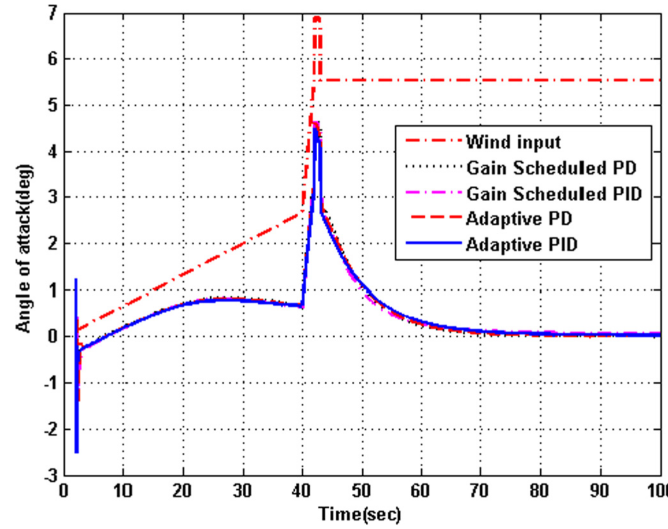

Fig. 13. Wind disturbance rejection of different controllers.

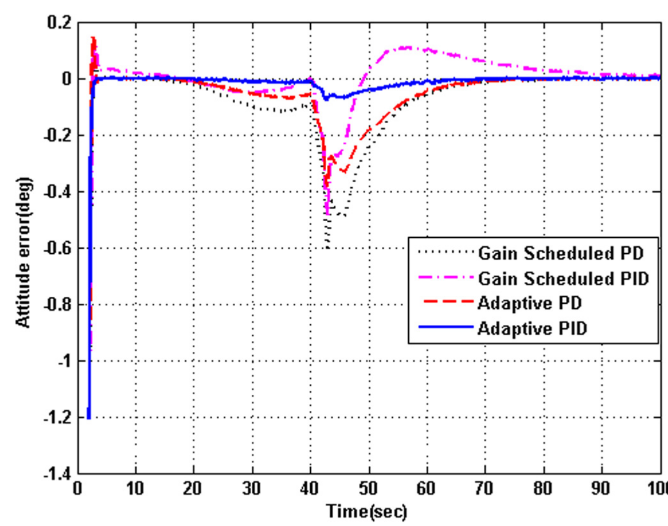

Fig. 14. Attitude errors build up due to wind for different controllers.

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Table 2
Comparison of the performance for different schemes with typical guidance commands under parameter perturbation.

Scheme	Tracking Error (deg)	Maximum Control Demand (deg)	Integral Absolute Square Error (deg)
Gain Scheduled PD Controller	0.3102	4 deg for initial capture 1.404 deg during high disturbance	3.2657
Gain Scheduled PID controller	0.09	4.16 deg for initial capture 1.0722 deg during high disturbance	0.6733
Adaptive PD Controller	0.1455	5.86 deg for initial capture 1.528 deg during high disturbance	0.3589
Adaptive PID Controller	0.06292	8 deg for initial capture 1.457 deg during high disturbance	0.0297

Table 3
Wind disturbance rejection characteristic for different schemes.

Scheme	Angle of Attack (deg)	Tracking Error (deg)	Maximum Control Demand (deg)	Integral Absolute Square Error (deg)
Gain Scheduled PD Controller	4.663	0.6103	4.04 deg for initial capture 2.36 deg during wind gust	2.2879
Gain Scheduled PID controller	4.631	0.4802	4.066 deg for initial capture 2.34 deg during wind gust	1.0804
Adaptive PD Controller	4.602	0.3804	5.82 deg for initial capture 2.20 deg during wind gust	0.8710
Adaptive PID Controller	4.49	0.078	8 deg for initial capture 2.142 deg during wind gust	0.1427

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